S&P GLOBAL & CRISIL CAMPUS HACKATHON

RiskLens

Explainable News-to-Risk Engine with Wholesale Banking Stress Testing

From unstructured text to auditable risk signals to portfolio stress tests.

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"SVB Financial stock plunges 60% as bank run fears intensify among tech clients"

Entity	**SIVB (Financials, US)**
Event type	**Credit Event**
Sentiment	**–0.96**
Impact score	**10.0**

Stress test triggered
–$142.0M on a $4.39B synthetic book

Auditable risk signals drive portfolio stress tests

THE PROBLEM

Thousands of headlines and posts a day. Most are noise. A few move credit risk.

Manual triage is slow, noisy and unauditable.

1

Ingest news and social text

Two source types, cleaned, deduplicated and linked to companies.

2

NLP engine scores every item

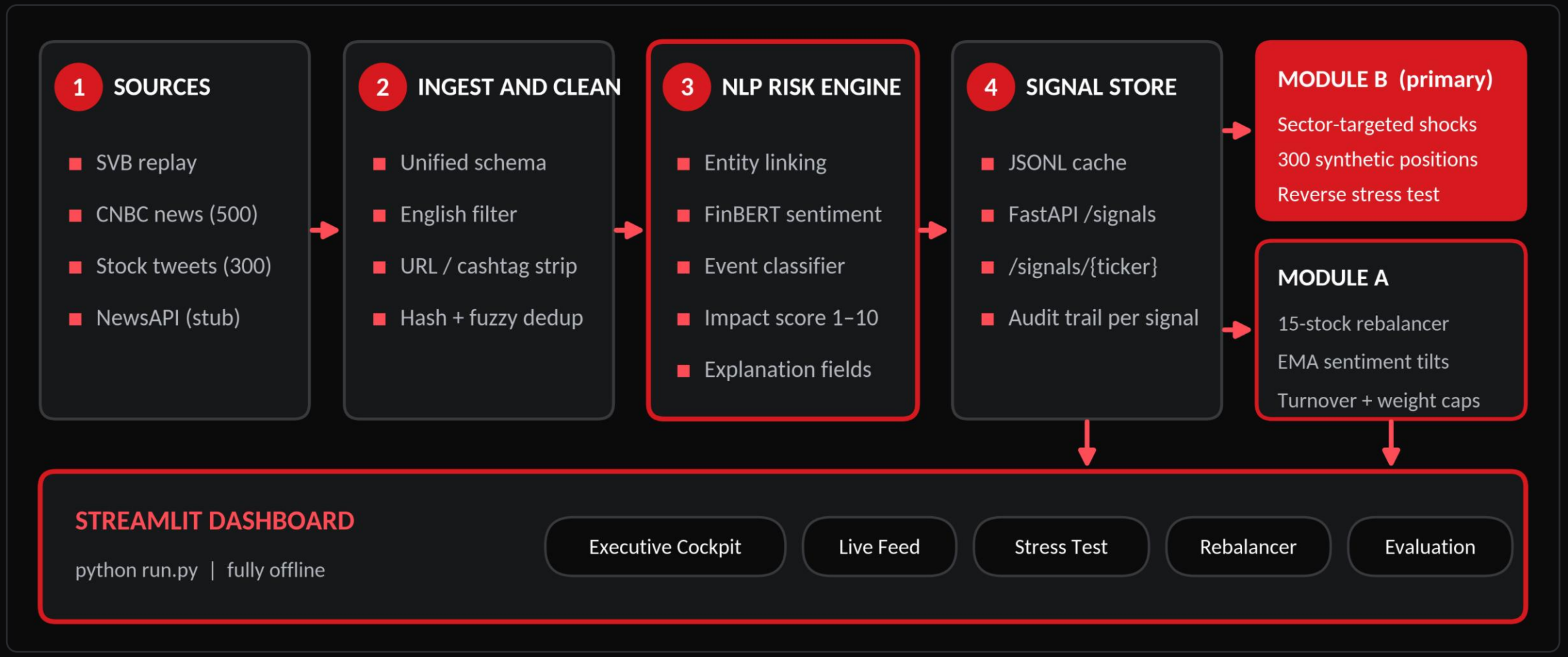
Sentiment, event type and an impact score from 1 to 10, with the factors shown.

3

High-impact events trigger action

A stress test on a wholesale bank portfolio and tilts on a stock index.

System design: from raw text to risk actions



Fully offline in replay mode: python run.py. Models run locally with no API keys.

Implementation highlights and why each choice

Component	Choice	Why
Sentiment	ProsusAI/finbert, fp16	Finance-domain model that fits a 4 GB GPU
Event class	MiniLM embeddings + logistic regression + keyword rules	Fast on CPU, with a transparent keyword fallback
Impact score	Severity × sentiment × credibility × novelty × corroboration	Every factor is shown on each signal
Stress test	Sector-targeted PD and spread shocks, 25% spillover	Credit-aware, not a blanket equity shock

EVALUATION BENCHMARKS (FROM THE DASHBOARD)

Empirical NLP & Classification Benchmarks

Model / Pipeline	Accuracy	Macro-F1	Adverse Recall	Role / Method
FinBERT (Argmax Mapping)	97.2%	0.9625	98.3%	Discrete argmax classification (academic ceiling)
FinBERT (Engine Net Threshold)	87.6%	0.8763	98.7%	Continuous net score P(pos) - P(neg), scales shocks in engine
VADER (Rule Baseline)	57.1%	0.4871	22.4%	Lexicon baseline (misses 77.6% of negative credit events)
MiniLM + LR Event Classifier	68.7%	0.6793	92.3% (Credit)	Stratified cross-validation out-of-fold generalization (n=67)

Offline replay, no API keys

39 unit tests

Signed reverse stress test

Key results: sentiment accuracy and the SVB stress test

SENTIMENT ACCURACY

97.2% FinBERT vs VADER 57.1%

THROUGHPUT

43.3 docs/sec on a GTX 1650 Ti

SIGNALS

820 processed and cached

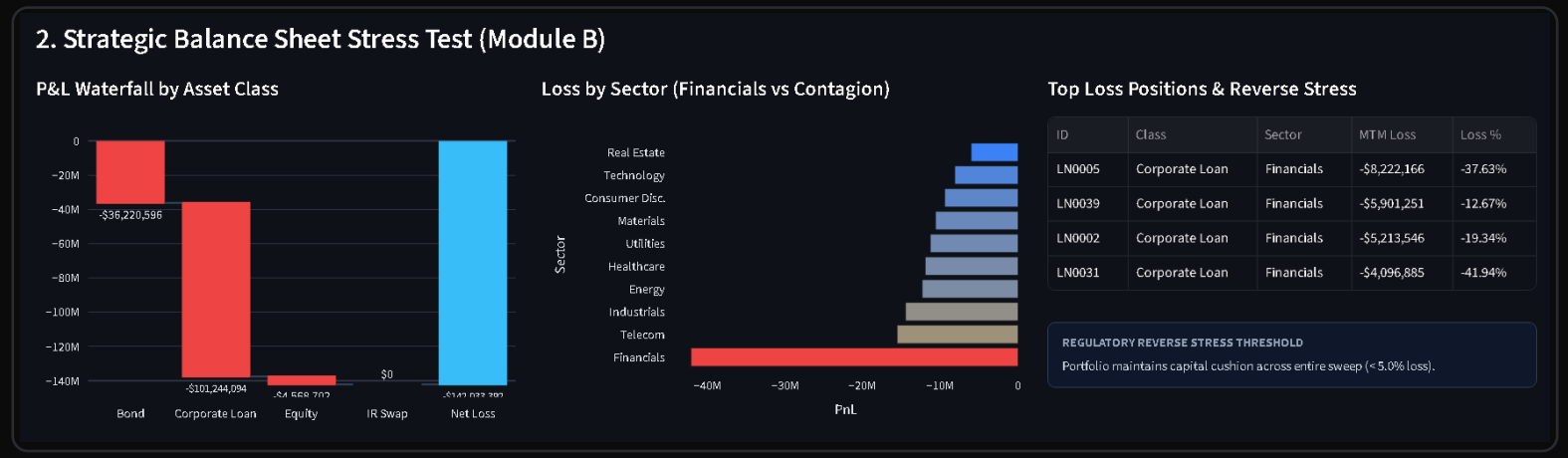
SVB TRIGGER | IMPACT 10.0

$4.39B → $4.25B

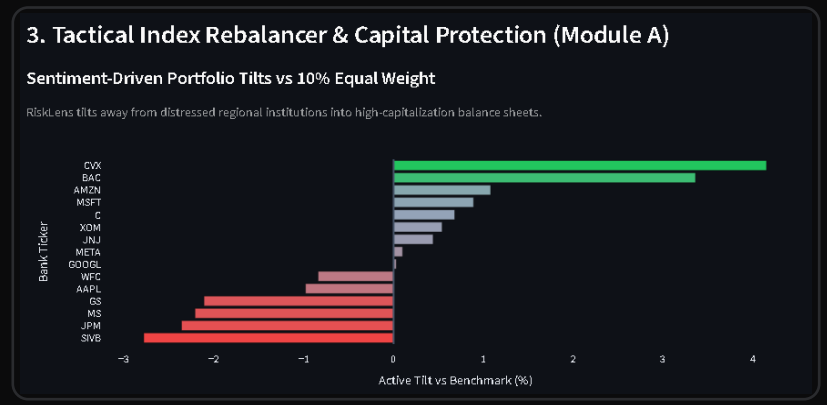
−$142.0M (−3.24%), Financials hit hardest

2.5% loss limit breached at impact 8.85

MODULE B: STRESS TEST (DASHBOARD)



MODULE A: REBALANCER TILTS (DASHBOARD)



What it shows

Distressed banks (SIVB, JPM, MS, GS) are tilted down; BAC and CVX up. Illustrative, driven by the replay, not a trading backtest.

FinBERT 97.2% uses the argmax class; the engine's threshold mapping scores 87.6%. The event classifier matches keyword rules (macro-F1 0.679, n=67).

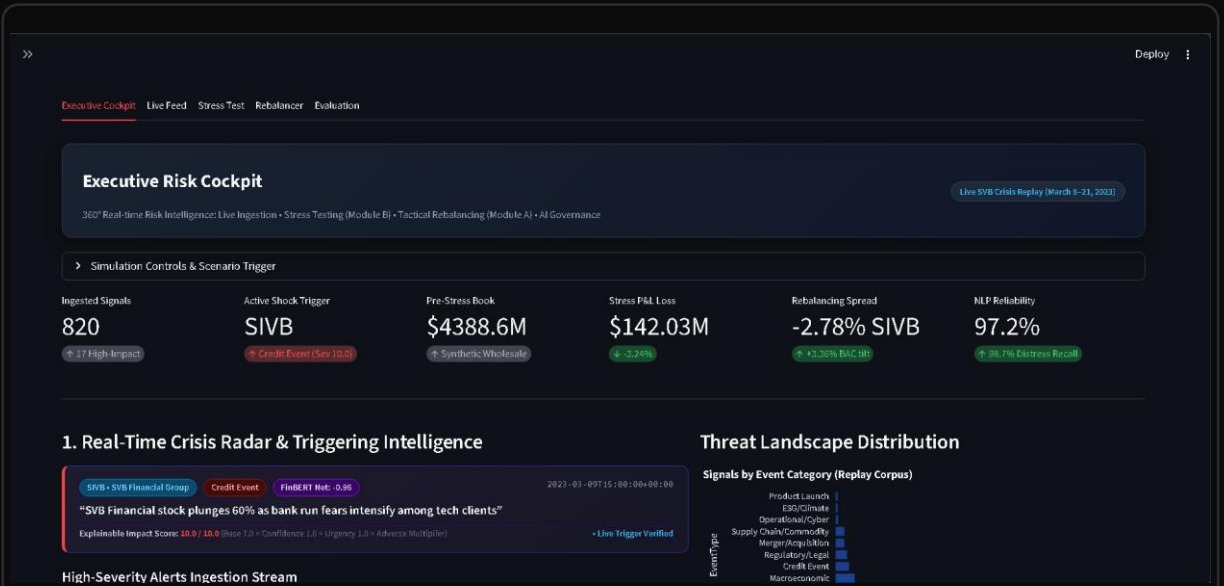
Domain impact for wholesale banking risk teams

Early warning
Surfaces credit-relevant events from raw text as they arrive.

Scenario linkage
Maps events to rating-bucket spreads and default-probability shocks.

Reverse stress testing
Finds the severity at which a chosen loss limit is breached.

Auditability
Every number traces to a headline, a source and the score factors.



Executive cockpit: signals, trigger, stress P&L, tilts and NLP reliability in one view.

Conceptually aligned with scenario analysis in ICAAP and supervisory stress testing.

Limitations and next steps

LIMITATIONS

- Synthetic portfolio and first-order pricing
- Impact severities are assumptions, not an event study
- Small labeled set (n=67): classifier acts like keyword rules
- SVB replay is a reconstruction from public reporting
- Sentiment is not causality

NEXT STEPS

- Event-study calibration of severities
- Larger labeled corpus and a fine-tuned classifier
- Live NewsAPI and GDELT feeds
- Richer pricing: CVA, FX and commodities